

Efficacy of Vesicular Stomatitis Virus-Ebola Virus Postexposure Treatment in Rhesus Macaques Infected With Ebola Virus Makona.

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The Ebola virus (EBOV) epidemic in West Africa increased the focus on vaccine development against this hemorrhagic fever-causing pathogen, and as a consequence human clinical trials for a few selected platforms were accelerated. One of these vaccines is vesicular stomatitis virus (VSV)-EBOV, also known as rVSV-ZEBOV, a fast-acting vaccine against EBOV and so far the only vaccine with reported efficacy against EBOV infections in humans in phase III clinical trials. In this study, we analyzed the potential of VSV-EBOV for postexposure treatment of rhesus macaques infected with EBOV-Makona. We treated groups of animals with 1 dose of VSV-EBOV either in a single injection at 1 or 24 hours after EBOV exposure or with 2 injections, half the dose at each time point; 1 control group received the same dose of the VSV-based Marburg virus vaccine at both time points; another group remained untreated. Although all untreated animals succumbed to EBOV infection, 33%-67% of the animals in each treatment group survived the infection, including the group treated with the VSV-based Marburg virus vaccine. This result suggests that protection from postexposure vaccination may be antigen unspecific and due rather to an early activation of the innate immune system. In conclusion, VSV-EBOV remains a potent and fast-acting prophylactic vaccine but demonstrates only limited efficacy in postexposure treatment.

Efficacy and effectiveness of an rVSV-vectored vaccine expressing Ebola surface glycoprotein: interim results from the Guinea ring vaccination cluster-randomised trial.

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A recombinant, replication-competent vesicular stomatitis virus-based vaccine expressing a surface glycoprotein of Zaire Ebolavirus (rVSV-ZEBOV) is a promising Ebola vaccine candidate. We report the results of an interim analysis of a trial of rVSV-ZEBOV in Guinea, west Africa. For this open-label, cluster-randomised ring vaccination trial, suspected cases of Ebola virus disease in Basse-Guinée (Guinea, west Africa) were independently ascertained by Ebola response teams as part of a national surveillance system. After laboratory confirmation of a new case, clusters of all contacts and contacts of contacts were defined and randomly allocated 1:1 to immediate vaccination or delayed (21 days later) vaccination with rVSV-ZEBOV (one dose of 2×10^7 plaque-forming units, administered intramuscularly in the deltoid muscle). Adults (age ≥ 18 years) who were not pregnant or breastfeeding were eligible for vaccination. Block randomisation was used, with randomly varying blocks, stratified by location (urban vs rural) and size of rings (≤ 20 vs > 20 individuals). The study is open label and masking of participants and field teams to the time of vaccination is not possible, but Ebola response teams and laboratory workers were unaware of allocation to immediate or delayed vaccination. Taking into account the incubation period of the virus of about 10 days, the prespecified primary outcome was laboratory-confirmed Ebola virus disease with onset of symptoms at least 10 days after randomisation. The primary analysis was per protocol and compared the incidence of Ebola virus disease in eligible and vaccinated individuals in immediate vaccination clusters with the incidence in eligible individuals in delayed vaccination clusters. This trial is registered with the Pan African Clinical Trials Registry, number PACTR201503001057193. Between April 1, 2015, and July 20, 2015, 90 clusters, with a total population of 7651 people were included in the planned interim analysis. 48 of these clusters (4123 people) were randomly assigned to immediate vaccination with rVSV-ZEBOV, and 42 clusters (3528 people) were randomly assigned to delayed vaccination with rVSV-ZEBOV. In the immediate vaccination group, there were no cases of Ebola

virus disease with symptom onset at least 10 days after randomisation, whereas in the delayed vaccination group there were 16 cases of Ebola virus disease from seven clusters, showing a vaccine efficacy of 100% (95% CI 74.7-100.0; $p=0.0036$). No new cases of Ebola virus disease were diagnosed in vaccinees from the immediate or delayed groups from 6 days post-vaccination. At the cluster level, with the inclusion of all eligible adults, vaccine effectiveness was 75.1% (95% CI -7.1 to 94.2; $p=0.1791$), and 76.3% (95% CI -15.5 to 95.1; $p=0.3351$) with the inclusion of everyone (eligible or not eligible for vaccination). 43 serious adverse events were reported; one serious adverse event was judged to be causally related to vaccination (a febrile episode in a vaccinated participant, which resolved without sequelae). Assessment of serious adverse events is ongoing. The results of this interim analysis indicate that rVSV-ZEBOV might be highly efficacious and safe in preventing Ebola virus disease, and is most likely effective at the population level when delivered during an Ebola virus disease outbreak via a ring vaccination strategy. WHO, with support from the Wellcome Trust (UK); Médecins Sans Frontières; the Norwegian Ministry of Foreign Affairs through the Research Council of Norway; and the Canadian Government through the Public Health Agency of Canada, Canadian Institutes of Health Research, International Development Research Centre, and Department of Foreign Affairs, Trade and Development.

Prevention and post-exposure management of occupational exposure to Ebola virus.

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There have been significant advances in the prevention and management of Ebola virus disease (EVD) caused by Zaire Ebola virus (ZEBOV), including the development of two effective vaccines, rVSV-ZEBOV and Ad26.ZEBOV/MVA-BN-Filo. In addition, ZEBOV monoclonal antibodies have become first-line therapy for EVD. However, the 2022-23 outbreak of Sudan Ebola virus (SUDV) in Uganda has highlighted the gap in current therapies and vaccines, whose efficacy is uncertain against non-ZEBOV species. Health-care and laboratory staff working in EVD treatment centres or Ebola virus diagnostic and research laboratories face unique risks relating to potential occupational exposure to Ebola viruses. Given the substantial morbidity and mortality associated with EVD, facilities should have strategies in place to manage occupational exposures, including consideration of post-exposure therapies. In this Review, we discuss currently available evidence for prevention and post-exposure prophylaxis of EVD, including therapies currently under evaluation for SUDV.

Rhabdoviruses as vaccine platforms for infectious disease and cancer.

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The family Rhabdoviridae (RV) comprises a large, genetically diverse collection of single-stranded, negative sense RNA viruses from the order Mononegavirales. Several RV members are being developed as live-attenuated vaccine vectors for the prevention or treatment of infectious disease and cancer. These include the prototype recombinant Vesicular Stomatitis Virus (rVSV) and the more recently developed recombinant Maraba Virus, both species within the genus Vesiculoviridae. A relatively strong safety profile in humans, robust immunogenicity and genetic malleability are key features that make the RV family attractive vaccine platforms. Currently, the rVSV vector is in preclinical development for vaccination against numerous high-priority infectious diseases, with clinical evaluation underway for HIV/AIDS and Ebola virus disease. Indeed, the success of the rVSV-ZEBOV vaccine during the 2014-15 Ebola virus outbreak in West Africa highlights the therapeutic potential of rVSV as a vaccine vector for

acute, life-threatening viral illnesses. The rVSV and rMaraba platforms are also being tested as 'oncolytic' cancer vaccines in a series of phase 1-2 clinical trials, after being proven effective at eliciting immune-mediated tumour regression in preclinical mouse models. In this review, we discuss the biological and genetic features that make RVs attractive vaccine platforms and the development and ongoing testing of rVSV and rMaraba strains as vaccine vectors for infectious disease and cancer.

Prior vaccination with rVSV-ZEBOV does not interfere with but improves efficacy of postexposure antibody treatment.

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A replication-competent vesicular stomatitis virus vaccine expressing the Ebola virus (EBOV) glycoprotein (GP) (rVSV-ZEBOV) was successfully used during the 2013-16 EBOV epidemic. Additionally, chimeric and human monoclonal antibodies (mAb) against the EBOV GP have shown promise in animals and humans when administered therapeutically. Uncertainty exists regarding the efficacy of postexposure antibody treatments in the event of a known exposure of a recent rVSV-ZEBOV vaccinee. Here, we model a worst-case scenario using rhesus monkeys vaccinated or unvaccinated with the rVSV-ZEBOV vaccine. We demonstrate that animals challenged with a uniformly lethal dose of EBOV one day following vaccination, and then treated with the anti-EBOV GP mAb MIL77 starting 3 days postexposure show no evidence of clinical illness and survive challenge. In contrast, animals receiving only vaccination or only mAb-based therapy become ill, with decreased survival compared to animals vaccinated and subsequently treated with MIL77. These results suggest that rVSV-ZEBOV augments immunotherapy.

Ring vaccination with rVSV-ZEBOV under expanded access in response to an outbreak of Ebola virus disease in Guinea, 2016: an operational and vaccine safety report.

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In March, 2016, a flare-up of Ebola virus disease was reported in Guinea, and in response ring vaccination with the unlicensed rVSV-ZEBOV vaccine was introduced under expanded access, the first time that an Ebola vaccine has been used in an outbreak setting outside a clinical trial. Here we describe the safety of rVSV-ZEBOV candidate vaccine and operational feasibility of ring vaccination as a reactive strategy in a resource-limited rural setting. Approval for expanded access and compassionate use was rapidly sought and obtained from relevant authorities. Vaccination teams and frozen vaccine were flown to the outbreak settings. Rings of contacts and contacts of contacts were defined and eligible individuals, who had given informed consent, were vaccinated and followed up for 21 days under good clinical practice conditions. Between March 17 and April 21, 2016, 1510 individuals were vaccinated in four rings in Guinea, including 303 individuals aged between 6 years and 17 years and 307 front-line workers. It took 10 days to vaccinate the first participant following the confirmation of the first case of Ebola virus disease. No secondary cases of Ebola virus disease occurred among the vaccinees. Adverse events following vaccination were reported in 47 (17%) 6-17 year olds (all mild) and 412 (36%) adults (individuals older than 18 years; 98% were mild). Children reported fewer arthralgia events than adults (one [$<1\%$] of 303 children vs 81 [7%] of 1207 adults). No severe vaccine-related adverse events were reported. The results show that a ring vaccination strategy can be rapidly and safely implemented at scale in response to Ebola virus disease outbreaks in rural settings. WHO, Gavi, and the World Food Programme.

Proteo-Genomic Analysis Identifies Two Major Sites of Vulnerability on Ebolavirus Glycoprotein for Neutralizing Antibodies in Convalescent Human Plasma.

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Three clinically relevant ebolaviruses - Ebola (EBOV), Bundibugyo (BDBV), and Sudan (SUDV) viruses, are responsible for severe disease and occasional deadly outbreaks in Africa. The largest Ebola virus disease (EVD) epidemic to date in 2013-2016 in West Africa highlighted the urgent need for countermeasures, leading to the development and FDA approval of the Ebola virus vaccine rVSV-ZEBOV (Ervebo®) in 2020 and two monoclonal antibody (mAb)-based therapeutics (Inmazeb® [atoltivimab, maftivimab, and odesivimab-ebgn] and Ebanga® [ansuvimab-zykl] in 2020. The humoral response plays an indispensable role in ebolavirus immunity, based on studies of mAbs isolated from the antibody genes in peripheral blood circulating ebolavirus-specific human memory B cells. However, antibodies in the body are not secreted by circulating memory B cells in the blood but rather principally by plasma cells in the bone marrow. Little is known about the protective polyclonal antibody responses in convalescent plasma. Here we exploited both single-cell antibody gene sequencing and proteomic sequencing approaches to assess the composition of the ebolavirus glycoprotein (GP)-reactive antibody repertoire in the plasma of an EVD survivor. We first identified 1,512 GP-specific mAb variable gene sequences from single cells in the memory B cell compartment. Using mass spectrometric analysis of the corresponding GP-specific plasma IgG, we found that only a portion of the large B cell antibody repertoire was represented in the plasma. Molecular and functional analysis of proteomics-identified mAbs revealed recognition of epitopes in three major antigenic sites - the GP head domain, the glycan cap, and the base region, with a high prevalence of neutralizing and protective mAb specificities that targeted the base and glycan cap regions on the GP. Polyclonal plasma antibodies from the survivor reacted broadly to EBOV, BDBV, and SUDV GP, while reactivity of the potentially neutralizing mAbs we identified was limited mostly to the homologous EBOV GP. Together these results reveal a restricted diversity of neutralizing humoral response in which mAbs targeting two antigenic sites on GP - glycan cap and base - play a principal role in plasma-antibody-mediated protective immunity against EVD.

rVSV-ZEBOV vaccination in people with pre-existing immunity to Ebolavirus: an open-label safety and immunogenicity study in Guinean communities affected by Ebola virus disease (l'essai proches).

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Zaire Ebolavirus disease (EVD) outbreaks can be controlled using rVSV-ZEBOV vaccination and other public health measures. People in high-risk areas may have pre-existing antibodies from asymptomatic Ebolavirus exposure that might affect response to rVSV-ZEBOV. Therefore, we assessed the impact pre-existing immunity had on post-vaccination IgG titre, virus neutralisation, and reactogenicity following vaccination. In this prospective cohort study, 2115 consenting close contacts ("proches") of EVD survivors were recruited. Proches were vaccinated with rVSV-ZEBOV and followed up for 28 days for safety and immunogenicity. Anti-GP IgG titre at baseline and day 28 was assessed by ELISA. Samples from a representative subset were evaluated using live virus neutralisation. Ten percent were seropositive at baseline. At day 28, IgG in baseline seronegative (GMT 0.106 IU/ml, 95% CI: 0.100 to 0.113) and seropositive (GMT 0.237 IU/ml, 0.210 to 0.267) participants significantly increased from baseline (both $p < 0.0001$). There was strong correlation between antibody titres and virus neutralisation

in day 28 samples (Spearman's rho 0.75). Vaccinees with baseline IgG antibodies against Zaire Ebolavirus had similar safety profiles to those without detectable antibodies (63.6% vs 66.1% adults experienced any adverse event; 49.1% vs 60.9% in children), with almost all adverse events graded as mild. No serious adverse events were attributed to vaccination. No EVD survivors tested positive for Ebolavirus by RT-PCR. These data add further evidence of rVSV-ZEBOV safety and immunogenicity, including in people with pre-existing antibodies from suspected natural ZEBOV infection whose state does not blunt rVSV-ZEBOV immune response. Pre-vaccination serological screening is not required.

Titration methods for rVSV-based vaccine manufacturing.

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The recombinant Vesicular Stomatitis Virus (rVSV) is an emerging platform for viral vector-based vaccines. Promising results have been reported in clinical trials for the rVSV-ZEBOV vaccine for Ebola virus disease prevention. In this study, we describe the titration tools elaborated to assess the titre of rVSV-ZEBOV productions. • A streamlined Median Tissue Culture Infectious Dose (TCID₅₀) assay to determine the infectious titer of this vaccine was established. • A digital polymerase chain reaction (dPCR) assay to assess the total number of viral particles present in cell-free culture supernatants of rVSV productions was developed. • These assays are used to titre rVSV-ZEBOV samples and characterize the ratio of total particles to infectious units for monitoring process robustness and product quality attributes and can be used to titre samples generated in the production of further rVSV vectors.

Production of rVSV-ZEBOV in serum-free suspension culture of HEK 293SF cells.

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Ebola virus disease is an urgent international priority. Promising results for several vaccine candidates have been reported in non-human primate studies and clinical trials with the most promising being the rVSV-ZEBOV vaccine. In this study, we sought to produce rVSV-ZEBOV in HEK 293SF cells in suspension and serum-free media. The purpose of this study was to establish a process using the HEK 293SF production platform, optimise the production titre, demonstrate scalability and the efficiency of the generated material to elicit an immune reaction in an animal model. Critical process parameters were evaluated to maximize production yield and process robustness and the following operating conditions: $1-2 \times 10^6$ cells/mL grown in HyClone HyCell TransFx-H media infected at an MOI of 0.001 with a temperature shift to 34 °C during the production phase and a harvest of the product after 48 h. Using these conditions, scalability in a 3.5 L controlled bioreactor was shown reaching a titre of 1.19×10^8 TCID₅₀/mL at the peak of production, the equivalent of 4165 doses of vaccine per litre. The produced virus was shown to be thermostable in the culture media and, when concentrated, purified and administered to mice, demonstrated the ability to induce a ZEBOV-specific immune response.